Lecture 4
Solutions

Homework 4: Branching out

Solutions to Homework 4

```
$ bash codeHints/L04/setup_hw4.sh
$ cd practice/hw4
$ git switch -c shopping
$ echo "milk" > shopping.txt
$ git add shopping.txt
$ git commit -m "Add shopping list"
$ echo "eggs" >> shopping.txt
$ git add shopping.txt
$ git commit -m "Add eggs"
$ git switch main
$ ls                          # no shopping.txt here
README.md  app.txt
$ echo "some notes" > notes.txt
$ git add notes.txt
$ git commit -m "Add notes"
$ git branch -m experimnt experiment
$ git branch hotfix main~1
$ git log --oneline --graph --all --decorate
* c6fac68 (HEAD -> main) Add notes
| * e7fa817 (shopping) Add eggs
| * 26c01df Add shopping list
|/
* 2e32674 (hotfix) Add app file
* 0a964a7 (experiment) Initial commit
$ printf 'branches=4\nbranch-is=pointer\n' > ../hw4_answers.txt
```

(The first two IDs, `0a964a7` and `2e32674`, are the same on your computer; the others depend on when you commit and will differ.)

4.1. `git switch -c shopping` creates and enters the branch at the `Add app file` commit, where `main` is at that moment. Creating it *before* the `Add notes` commit matters: the fork point is where the two lines of work separate.

4.2. Both commits go onto `shopping` because that is the branch `HEAD` points at. Check with `git status` (first line) before each commit.

4.3. After `git switch main` your files change: `shopping.txt` disappears from the folder. It is not lost; it lives in the commits of `shopping`. The `Add notes` commit moves `main` only. `shopping` stays where it was.

4.4. `git branch -m old new` renames. The branch still points at the first commit, so the checker's "experiment is the first commit" test passes.

4.5. `git branch hotfix main~1` creates the label at the parent of the newest `main` commit, which is `Add app file`, without switching. (`git switch -c` would also have created it, but you would then have to switch back.)

4.6. The four branches are `experiment`, `hotfix`, `main` and `shopping`. The word is **pointer**; "label" and "reference" are also accepted.

Common mistakes.

- Making the `shopping` commits while still on `main` (forgetting to switch). The checker then finds main with too many commits. Use `setup_hw4.sh --reset` and start again, watching the first line of `git status`.
- Creating `shopping` *after* the `Add notes` commit, so it forks from the wrong place.
- Using `git branch -D` or deleting `experimnt` instead of renaming it: the commit is still there but the branch name must be `experiment`.
- Ending the exercise on another branch: the last task asks you to be on `main`.